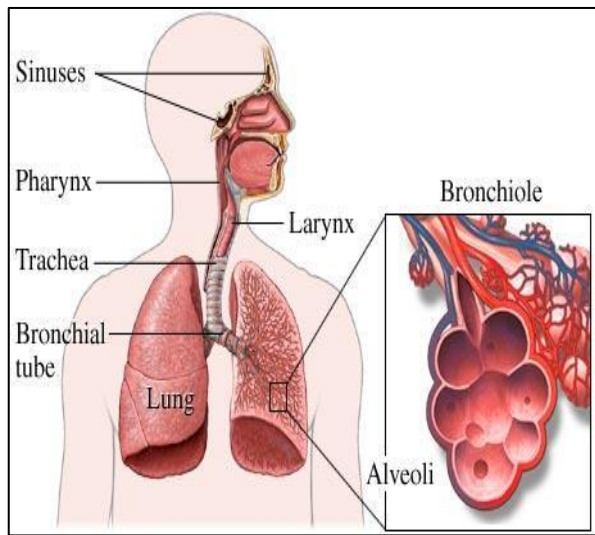




RESPIRATORY SYSTEM

The respiratory system consists of organs that help you breathe. *Breathing* is a process that allows your body to take oxygen and release carbon dioxide.

The Parts and Functions of the Respiratory System

Nasal Cavity

- It is where air enters the nostril.
- It has hairlike structures called **cilia** that filter dust and pathogen from the air.
- It has mucous membrane that moistens the air before it reaches the lungs.

Pharynx

- It is where the warm and moistened air passes.
- It is a muscular tube in which food, water, and air pass through.
- A flap tissue called the **epiglottis** prevents food and water entering the larynx. Larynx is commonly called as voice box.

Trachea or Windpipe

- Below the larynx is trachea.
- It is a hollow muscular tubular structure.
- Air passes through the larynx and trachea on its way to the lungs.

Bronchi and Bronchioles

- **Trachea** divides into passageways, called bronchi, leading to the left and right lungs.
- **Bronchi** transport air to and from the lungs.
- The **cartilage** prevents the bronchi from collapsing when you inhale and exhale.
- **Mucosal lining** produces mucous which trap foreign particles before it enters the lungs.
- **Bronchioles** are the branches of bronchi.
- **Alveoli** are tiny air sacs at bronchioles end where gas exchange occurs.
- **Capillaries** are blood vessels that surround alveoli.
- The oxygen diffuses in the alveoli into the blood while carbon dioxide diffuses out of the blood into the alveoli, then exhaled out of the body.

Lungs

- The lungs are the main organs of the respiratory system.
- In the lungs, oxygen is taken in and transported to the different parts of the body and carbon dioxide is expelled out back to the atmosphere.

Mechanics of Breathing

- Inhalation and exhalation are phases of respiration.
- **Inhalation** happens when oxygen from the air enters the lungs. As you inhale, you can observe that your chest moves outward and upward to expand as a result of the contraction of rib muscles. Furthermore, the diaphragm contracts and flattens as you inhale.
- During **exhalation**, air is forced out of the lungs. The diaphragm and rib muscles relax and return to its original shape. This makes the chest move inward or downward causing the carbon dioxide to rush out of the body.